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Intersect Milestone Acceptance Form

Milestone Acceptance Form (MAF)

Please complete this form once you have completed a milestone under your contract.

If you have an on-chain contract, please upload the completed form to the UI dashboard under the relevant milestone for review by Intersect's Delivery Assurance Team.

If you do not have an on-chain contract, please e-mail to billing@intersectmbo.org under the relevant milestone for review by Intersect's Delivery Assurance Team.

Project Name	*LambdaBuffers*
Project Number	*OSC-26-0040*
Milestone Name(s) & Number(s)	*Milestone 1 – Dependency Updates*
Milestone Description	
1. Update the LambdaBuffers Nix dependency set to current compatible versions on the staging branch. 2. Repair and validate the LambdaBuffers Nix shell so it launches successfully without errors. 3. Ensure the staging-branch Nix shell exposes all dependencies required to build and develop LambdaBuffers at the updated versions. 4. Record the dependency and Nix-environment changes in CHANGELOG.md. 5. Provide a brief, 2–5 minute video overview of the Milestone activities.	
Acceptance Criteria Met	

Acceptance Criteria 1

Criteria: All dependencies for LambdaBuffers are up to their current versions at the time of this Milestone being submitted on the staging branch.

Completion Statement: All Cardano-related dependencies (Plutus/CHaP, Plutarch, cardano-transaction-lib, the Rust plutus-ledger-api crate) were updated to current, PV11-compatible versions on staging. The GHC toolchain was bumped from 9.6.7 to 9.12.1 to satisfy the combined requirements of Plutarch 1.14 and plutus-tx-plugin 1.65.

Verifiable Evidence:

• Pull request: https://github.com/mlabs-haskell/lambda-buffers/pull/297

• Head commit: https://github.com/mlabs-haskell/lambda-buffers/commit/bfa9bcdac3290a503b426cd850d86dbbfee86a81

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• Plutarch pin (flake.lock, plutarch input → 1.14.0):
https://github.com/mlabs-haskell/lambda-buffers/blob/bfa9bcdac3290a503b426cd850d86dbbfee86a81/flake.lock#L3940-L3963

• CHaP pin (flake.lock, plutus 1.65-capable rev):
https://github.com/mlabs-haskell/lambda-buffers/blob/bfa9bcdac3290a503b426cd850d86dbbfee86a81/flake.lock#L354-L364

Acceptance Criteria 2

Criteria: LambdaBuffers' Nix shell can be brought up without errors on the staging branch.

Completion Statement: The flake evaluates and every Nix dev shell launches cleanly, confirmed by a green Hercules CI evaluation and full build/check run against the head commit.

Verifiable Evidence:

• Hercules CI evaluation (SUCCESS):
https://hercules-ci.com/github/mlabs-haskell/lambda-buffers/jobs/3859

• Hercules CI build/check (SUCCESS):
https://hercules-ci.com/github/mlabs-haskell/lambda-buffers/jobs/3860

Acceptance Criteria 3

Criteria: The Nix shell for LambdaBuffers' staging branch makes all dependencies required to build and develop LambdaBuffers at the updated versions.

Completion Statement: Every CHaP-enabled Haskell project (lbr-plutus, lbr-plutarch, lbr-plutustx, the lbt-plutus-* testsuites, docs/plutarch, docs/plutustx, extras/dev-shells/plutus-haskell) is pinned to a uniform plutus-core/plutus-ledger-api/plutus-tx/plutus-tx-plugin ==1.65.0.0 constraint, and the shared settings.nix module pins the compiler to GHC 9.12.1 for all dev shells built from it.

Verifiable Evidence:

• Compiler pin (settings.nix):
https://github.com/mlabs-haskell/lambda-buffers/blob/bfa9bcdac3290a503b426cd850d86dbbfee86a81/settings.nix#L72-L79

• Uniform plutus version constraint (lbr-plutus/cabal.project):
https://github.com/mlabs-haskell/lambda-buffers/blob/bfa9bcdac3290a503b426cd850d86dbbfee86a81/runtimes/haskell/lbr-plutus/cabal.project#L7-L10

• CHANGELOG.md dependency table:
https://github.com/mlabs-haskell/lambda-buffers/blob/bfa9bcdac3290a503b426cd850d86dbbf

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ee86a81/CHANGELOG.md#lambdabuffers-v210

Acceptance Criteria 4

Criteria: CHANGELOG.md on the staging branch contains at least one entry describing the changes made as part of this Milestone.

Completion Statement: A LambdaBuffers v2.1.0 entry was added to CHANGELOG.md, describing the van Rossem/PV11 hard fork context and every dependency and Nix-environment change made (Plutus/CHaP, Plutarch, GHC, cardano-transaction-lib, purescript-cardano-package-set, and the Rust plutus-ledger-api crate), and confirming no LambdaBuffers API or serialization changes were required.

Verifiable Evidence:

• https://github.com/mlabs-haskell/lambda-buffers/blob/bfa9bcdac3290a503b426cd850d86dbbfee86a81/CHANGELOG.md#lambdabuffers-v210

Acceptance Criteria 5

Criteria: Video overview demonstrates Nix updates and successful setup, dependency updates, and CHANGELOG.md updates.

Completion Statement: A brief 2–5 minute video overview of the Milestone 1 activities has been recorded, demonstrating the Nix dependency updates, the successful Nix shell setup, and the CHANGELOG.md updates.

Verifiable Evidence:

• Video: https://drive.google.com/file/d/1FaDhIw2heYT807yMwinfDAuLebS707uQ/view?usp=drive_link

Quality & Testing

Hercules CI ran a full evaluation and build/check pass against the submitted head commit; both are green (see Acceptance Criteria 1 and 2 evidence). Per CHANGELOG.md, the van Rossem hard fork is an intra-era Conway upgrade with no new ledger era or Plutus language version, and no LambdaBuffers API or serialization changes were required — all serialization formats remain backward compatible across the PureScript/Rust/Haskell/TypeScript/Plutarch/PlutusTx targets listed in the changelog's Schemas section.

Public Communication Summary

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LambdaBuffers is MLabs' code-generation toolkit that compiles language-agnostic type schemas into native serialization code across PureScript, Rust, Haskell, TypeScript, Plutus Tx, and Plutarch — a foundational piece of Cardano developer tooling.

This milestone is part of the broader effort to keep the Cardano open-source tooling ecosystem ready for the Van Rossem hard fork. Before new features can ship, the underlying dependencies and developer environments must be brought up to the latest Protocol Version 11 (PV11)-capable versions.

For Milestone 1, MLabs updated LambdaBuffers' entire Nix dependency set on the staging branch — including Plutus/CHaP, Plutarch, cardano-transaction-lib, and the Rust plutus-ledger-api crate — and upgraded the GHC compiler from 9.6.7 to 9.12.1. The Nix development shell now launches cleanly and exposes every dependency at its current version, confirmed by a green CI evaluation and build/check, with the changes recorded in CHANGELOG.md.

For the Cardano community, this keeps LambdaBuffers and everything that depends on it building and testing against the Van Rossem-compatible dependency set, helping ensure the wider tooling ecosystem stays stable and current as the network upgrades.

Announced on X: https://x.com/MLabs10/status/2092643673590636829

Announced on LinkedIn: https://www.linkedin.com/feed/update/urn:li:share:7498418188833931265

Signature Authorization

Role	Name	Signature	Date of Signature
The person responsible for producing the work, e.g. Project Lead / Technical Owner			
Project Manager	*Ben Hart*	Signed by: Ben Hart 4E496D17D87E40B...	9/9/2026

Third Party Assurer Comments (OSC Chair)
The project advances as expected.

Third Party Assurer Signature

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OSC Chair	*Sebastian Pabon*	Sebastian Pabon	*27/08/2026*

Signed on behalf of the Delivery Assurance Team, Responsible for overseeing the administrative process (Delivery Assurance Process and Best Practice Policy)

Intersect Delivery Assurance			
Intersect Delivery Assurance Partner	Operational Services	Signed by: Operational Services 01F12670ABAC4E3...	9/10/2026